Using State-Space Models to enhance Signal-Generation in Statistical Arbitrage Trading Strategies

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1 The Markov Property

The Markov property is the probabilistic foundation of all State-Space models. It formalizes the idea that, given the present, the future is conditionally independent of the past: the knowledge of the system's current state contains all the relevant information, and knowing the previous history adds nothing to the prediction of future states.

First Order Markov Property

A stochastic process $\{X_t\}_{t\geq 0}$ satisfies the *first-order Markov property* if, for every t and for every measurable set $A \in \mathcal{B}(\mathbb{R})$[1]:

$$\mathbb{P}(X_{t+1} \in A \mid \mathcal{F}_t) = \mathbb{P}(X_{t+1} \in A \mid X_t) \tag{1}$$

where $\mathcal{F}_t = \sigma(X_0, X_1, \ldots, X_t)$ is the natural filtration generated by the entire history of the process up to time t.
Condition (1) states that the conditional distribution of X_{t+1}, once conditioned on X_t, does not depend on $X_{t-1}, X_{t-2}, \ldots, X_0$. In terms of conditional densities, the property is equivalent to requiring:

$$p(x_{t+1} \mid x_t, x_{t-1}, \ldots, x_0) = p(x_{t+1} \mid x_t) \tag{2}$$

for almost every realization of the sequence. The Markov property is not as restrictive an assumption as it might seem at first glance. Any higher-order stochastic process can be rewritten as a first-order Markov process through an appropriate redefinition of the state space. An AR(p), for instance, is not Markovian in Y_t but it is in the augmented vector $(Y_t, Y_{t-1}, \ldots, Y_{t-p+1})'$. This observation is exactly analogous to the companion form of the VAR: the VAR(P) is not first-order in Y_t, but becomes a VAR(1) in $Z_t = (Y_t', Y_{t-1}', \ldots, Y_{t-P+1}')'$.

Transition Kernel and Chapman-Kolmogorov Equation

For continuous state-space processes $\mathcal{X} \subseteq \mathbb{R}^n$, the Markovian dynamics are completely characterized by the *transition kernel*:

$$p(x_{t+1} \mid x_t) = \frac{\partial}{\partial x_{t+1}} \mathbb{P}(X_{t+1} \leq x_{t+1} \mid X_t = x_t) \tag{3}$$

which assigns to each current state x_t a probability distribution over the subsequent states. The transition kernel is the fundamental object that encodes all the dynamics of the process. Knowing $p(x_{t+1} \mid x_t)$ is equivalent to knowing the laws of motion of the system. The consistency of the kernel over multiple horizons is guaranteed by the Chapman-Kolmogorov equation:

$$p(x_{t+h} \mid x_t) = \int_{\mathcal{X}} p(x_{t+h} \mid x_{t+h-1})\, p(x_{t+h-1} \mid x_t)\, dx_{t+h-1} \tag{4}$$

Equation (4) states that the probability of moving from state x_t to state x_{t+h} in h steps is obtained by integrating over all possible intermediate states x_{t+h-1}. The h-step future is the composition of the one-step future applied recursively. For discrete state-space processes $\mathcal{S} = \{1, 2, \ldots, K\}$, the transition kernel becomes the transition matrix $\mathbf{P}$ with entries $p_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$ and the Chapman-Kolmogorov equation reduces to matrix multiplication: the transition probability in h steps is the (i, j) element of the matrix $\mathbf{P}^h$.

[1] $\mathcal{B}$ denotes the Borel sigma-algebra

Stationarity and Stationary Distribution

A Markov chain is called time-homogeneous if the transition kernel does not depend on t: $p(x_{t+1} \mid x_t) = p(x_1 \mid x_0)$ for every t. Under this assumption (which we will hold throughout), the dynamics of the process are completely characterized by the kernel alone. A distribution π^* is called stationary for the Markov chain if:

$$\pi^*(A) = \int_{\mathcal{X}} p(x' \mid x)\, \pi^*(dx) \quad \forall\, A \in \mathcal{B}(\mathbb{R}) \tag{5}$$

or, in the discrete case, $\boldsymbol{\pi}^* = \boldsymbol{\pi}^* \mathbf{P}$. The stationary distribution is the left eigenvector of $\mathbf{P}$ associated with the unit eigenvalue. Under ergodicity conditions, the distribution of the chain converges to the stationary distribution regardless of the initial condition: $\Pr(X_t = j) \to \pi_j^*$ as $t \to \infty$. In the context of HMMs, π_k^* represents the expected proportion of time the system spends in regime k in the long run.

StatArb Implications

The Markov property has a fundamental consequence for pairs trading. If the latent state vector $\boldsymbol{\alpha}_t = (\alpha_t, \beta_t, \sigma_t^2, s_t)'$ (which includes the intercept, hedge ratio, conditional volatility, and market regime) satisfies the Markov property, then the optimal forecast of the spread at time $t+1$ does not require the memory of the entire price history but only the current estimate of the latent state. This property is the principle on which the Kalman Filter recursion and the HMM forward algorithm are founded. Both update the latent state estimate using a single iteration that combines the previous estimate with the new observation, without ever directly accessing the complete history.

2 Limits of Static Cointegration and Dynamic OLS

Before introducing state-space models, it is necessary to motivate their use by analyzing the limitations of static methodologies.

The Issue with the Static Hedge Ratio

The OLS estimate of the hedge ratio in the cointegrating regression is:

$$Y_{1,t} = \alpha + \beta\, Y_{2,t} + \varepsilon_t \tag{6}$$

with estimator:

$$\hat{\beta}_{\text{OLS}} = \frac{\sum_{t=1}^{T} (Y_{2,t} - \bar{Y}_2)(Y_{1,t} - \bar{Y}_1)}{\sum_{t=1}^{T} (Y_{2,t} - \bar{Y}_2)^2} \tag{7}$$

This estimator is super-consistent in the presence of cointegration [**engle_granger1987**]: it converges to the true parameter β at rate T instead of $\sqrt{T}$. Super-consistency is a remarkable property, but it has a structural problem: $\hat{\beta}_{\text{OLS}}$ estimates the temporal mean of β_t over the entire sample, not its current value at time T.
Let the true hedge ratio be a time-varying process:

$$\beta_t = \beta + \delta_t \tag{8}$$

where β is the constant component and δ_t is the time-varying component with $\mathbb{E}[\delta_t] = 0$. The DGP is:

$$Y_{1,t} = \alpha_t + \beta_t\, Y_{2,t} + u_t = \alpha_t + (\beta + \delta_t)\, Y_{2,t} + u_t \tag{9}$$

Estimating the static regression (6) on these data, the OLS residual is:

$$\begin{aligned} \hat{\varepsilon}_t &= Y_{1,t} - \hat{\alpha} - \hat{\beta}_{\text{OLS}}\, Y_{2,t} \\ &= \underbrace{(Y_{1,t} - \alpha_t - \beta_t\, Y_{2,t})}_{u_t \sim I(0)} + \underbrace{(\alpha_t - \hat{\alpha})}_{I(0) \text{ or } I(1)} + \underbrace{(\beta_t - \hat{\beta}_{\text{OLS}})\, Y_{2,t}}_{\delta_t Y_{2,t}} \end{aligned} \tag{10}$$

The critical term is $\delta_t Y_{2,t}$. Since $Y_{2,t} \sim I(1)$ and δ_t is generally correlated with the movements of $Y_{2,t}$ (the parameter drift is not independent of the asset being hedged), this term is $I(1)$. The OLS spread $\hat{\varepsilon}_t$ is therefore non-stationary, even though the true cointegrating relationship is correct. The ADF test applied to $\hat{\varepsilon}_t$ will tend not to reject the unit root, leading to the false conclusion that cointegration has broken down.

DOLS Limitations

Dynamic OLS (DOLS), proposed by Stock and Watson [**stock_watson1993**], corrects the bias of the OLS estimator by including leads and lags of the first differences of the regressor:

$$Y_{1,t} = \alpha + \beta\, Y_{2,t} + \sum_{j=-q}^{q} \delta_j\, \Delta Y_{2,t-j} + u_t \tag{11}$$

The terms $\Delta Y_{2,t-j}$ eliminate the part of the endogeneity resulting from the correlation between $Y_{2,t}$ and the shocks u_t (correlation caused by the common stochastic trend), and the terms with $j > 0$ (leads) capture any anticipatory correlation. The DOLS estimator of β has a mixed normal asymptotic distribution and converges faster than OLS in finite samples. However, DOLS remains a fixed-parameter model. β is still estimated as a constant over the entire sample $t = 1, \ldots, T$. Equation (11) locks the coefficient to a constant and therefore remains inadequate in the presence of structural breaks, regime changes, or continuous parameter drift. The problem formalized in equation (10) fully applies to DOLS: correcting the bias does not solve the problem of parameter instability.

Parametric Stability Test

Parameter stability in the cointegrating regression can be tested using the CUSUM test by Brown, Durbin, and Evans [**brown_durbin_evans1975**] or the Nyblom tests [**nyblom1989**]. The Nyblom test examines the null hypothesis of constant parameters against the alternative that the coefficients follow a random walk:

$$H_0 : \beta_t = \beta \quad \forall t \qquad \text{vs} \qquad H_1 : \beta_t = \beta_{t-1} + \eta_t, \quad \eta_t \sim \text{i.i.d.}(0, \sigma_\eta^2) \tag{12}$$

The Nyblom statistic for parameter β is:

$$L = \frac{1}{\hat{\sigma}_u^2 T^2} \sum_{t=1}^{T} \left(\sum_{s=1}^{t} \hat{u}_s\, Y_{2,s} \right)^2 \tag{13}$$

High values of L (above tabulated critical values) lead to the rejection of H_0 and indicate that the constant parameter assumption is statistically untenable. The parameter σ_η^2 under the alternative H_1 is the variance of the state process q_β in the Kalman Filter model, establishing a direct link between the Nyblom test and filter calibration.

Trading Signals Implications

These theoretical limitations translate into operational consequences for trading. The z-score of the classic strategy is:

$$z_t^{\text{OLS}} = \frac{Y_{1,t} - \hat{\alpha} - \hat{\beta}_{\text{OLS}}\, Y_{2,t}}{\hat{\sigma}_\varepsilon} \tag{14}$$

where $\hat{\sigma}_\varepsilon$ is the sample standard deviation of the OLS residuals over the entire sample. This z-score is biased in three ways:

1. **Numerator bias:** the residual includes the non-stationary term $\delta_t Y_{2,t}$, as shown in (10). During periods when β_t diverges from $\hat{\beta}_{\text{OLS}}$, the numerator takes on extreme values not due to a normal mean-reverting deviation of the spread but due to the incorrect estimation of the hedge ratio.

2. **Denominator bias:** $\hat{\sigma}_\varepsilon$ is estimated over the entire sample and reflects historical average volatility. During high volatility periods, $\hat{\sigma}_\varepsilon$ underestimates the current standard deviation, artificially inflating the z-score and multiplying false entry signals precisely when market risk is highest.

3. **Lack of regime adaptation:** the static z-score does not distinguish between a wide spread deviation caused by a genuine mean-reversion opportunity and a deviation caused by a structural break in cointegration. In both cases, z_t^{OLS} exceeds the threshold z^* and generates an entry signal, but only in the first case should the position converge back to equilibrium.

The State-Space Models framework, developed in the following sections, solves all three problems: the Kalman Filter corrects the numerator by dynamically updating $\hat{\beta}_{t|t}$ and the variance of the innovation f_t corrects the denominator, while the HMM solves the third problem by filtering regimes and conditioning signal execution on cointegration stability.

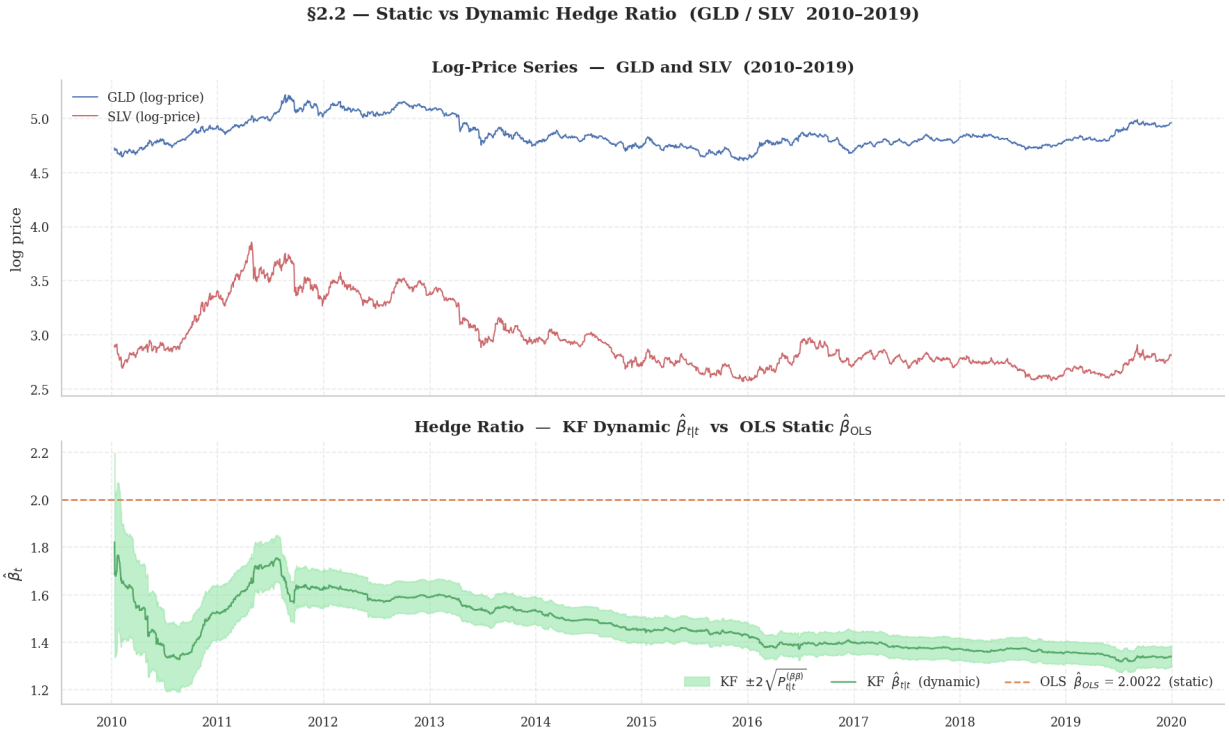


Figure 1: Top panel: time series of GLD and SLV log-prices (2010–2019). Bottom panel: comparison between the dynamic hedge ratio $\hat{\beta}_{t|t}$ estimated with the Kalman Filter (green line, with confidence band $\pm 2\sqrt{P_{t|t}^{(\beta\beta)}}$) and the static OLS estimate $\hat{\beta}_{\text{OLS}}$ (orange dashed line). The KF hedge ratio varies significantly over time, moving from values close to 2 in 2010 to roughly 1.35 in 2019, highlighting the structural inadequacy of the constant parameter assumption implicit in OLS.

3 State-Space Representation

The state-space representation is the formal framework that separates observable variables from latent ones and describes the law of motion of the latter over time. It was originally developed in control systems engineering, subsequently adopted in econometrics by Harvey [**harvey1989**], and formalized by Durbin and Koopman [**durbin_koopman2012**]. The framework is quite general: AR, ARMA, VECM, GARCH models, unobserved components models, and local trend models are all special cases of the state-space representation.

Linear Gaussian State-Space Model

A linear Gaussian state-space system is defined by the equations:

$$\boldsymbol{y}_t = \mathbf{Z}_t\,\boldsymbol{\alpha}_t + \mathbf{d}_t + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0},\, \mathbf{H}_t) \tag{15}$$

$$\boldsymbol{\alpha}_{t+1} = \mathbf{T}_t\,\boldsymbol{\alpha}_t + \mathbf{c}_t + \mathbf{R}_t\,\boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0},\, \mathbf{Q}_t) \tag{16}$$

for $t = 1, \ldots, T$, where $\boldsymbol{y}_t \in \mathbb{R}^p$ is the vector of observed variables, $\boldsymbol{\alpha}_t \in \mathbb{R}^m$ is the latent state vector, $\mathbf{Z}_t \in \mathbb{R}^{p\times m}$ is the *design matrix*, $\mathbf{T}_t \in \mathbb{R}^{m\times m}$ is the *transition matrix*, $\mathbf{R}_t \in \mathbb{R}^{m\times g}$ is the *selection matrix*, $\mathbf{H}_t \in \mathbb{R}^{p\times p}$ and $\mathbf{Q}_t \in \mathbb{R}^{g\times g}$ are the covariance matrices of the innovations, and $\mathbf{d}_t$, $\mathbf{c}_t$ are vectors of deterministic components. Zero correlation between the noise terms is assumed: $\mathbb{E}[\boldsymbol{\varepsilon}_t\boldsymbol{\eta}_s'] = \mathbf{0}$ for every t, s.

Components Interpretation

The **observation equation** (15) describes how the unobservable latent state $\boldsymbol{\alpha}_t$ generates the observations $\boldsymbol{y}_t$. Matrix $\mathbf{Z}_t$ determines the relationship between state and observation; $\boldsymbol{\varepsilon}_t$ is the measurement noise that captures sampling errors, rounding, and idiosyncratic components.
The **transition equation** (16) describes the law of motion of the latent state vector: how $\boldsymbol{\alpha}_t$ evolves over time. Matrix $\mathbf{T}_t$ describes state persistence, playing the role of autoregressive coefficients in a VAR. Matrix $\mathbf{R}_t$ selects which elements of the process noise $\boldsymbol{\eta}_t$ affect which components of the state. Matrix $\mathbf{Q}_t$ controls the speed of state variation: a small $\mathbf{Q}_t$ implies a slowly varying state (stable parameters), while a large $\mathbf{Q}_t$ allows for rapid variations.

Markovian Structure

A fundamental aspect of the state-space model is that the state $\boldsymbol{\alpha}_t$ is a first-order Markov process: given the transition equation (16) and Gaussianity assumptions, the distribution of $\boldsymbol{\alpha}_{t+1}$ depends only on $\boldsymbol{\alpha}_t$ and not on prior history. This implies that the joint distribution of the entire system $(\boldsymbol{y}_{1:T}, \boldsymbol{\alpha}_{1:T})$ factorizes as:

$$p(\boldsymbol{y}_{1:T}, \boldsymbol{\alpha}_{1:T}) = p(\boldsymbol{\alpha}_1) \prod_{t=1}^{T} p(\boldsymbol{y}_t \mid \boldsymbol{\alpha}_t) \prod_{t=1}^{T-1} p(\boldsymbol{\alpha}_{t+1} \mid \boldsymbol{\alpha}_t) \tag{17}$$

This factorization provides the structure that makes recursive inference possible via the Kalman Filter. The quantities $p(\boldsymbol{y}_t \mid \boldsymbol{\alpha}_t)$ and $p(\boldsymbol{\alpha}_{t+1} \mid \boldsymbol{\alpha}_t)$ are directly readable from equations (15)–(16).

Initialization

To start the recursion, the initial distribution of the state must be specified:

$$\boldsymbol{\alpha}_1 \sim \mathcal{N}(\boldsymbol{a}_1, \mathbf{P}_1) \tag{18}$$

where $\boldsymbol{a}_1 = \mathbb{E}[\boldsymbol{\alpha}_1]$ is the prior mean and $\mathbf{P}_1 = \mathrm{Var}(\boldsymbol{\alpha}_1)$ is the initial covariance. For stationary states like an AR(1) with $|\phi| < 1$, $\mathbf{P}_1$ can be set to the unconditional covariance of the process. For non-stationary states like a random walk (the case for the dynamic hedge ratio), *diffuse initialization* is used: $\mathbf{P}_1 = \kappa\mathbf{I}$ with $\kappa \to \infty$, which equates to imposing a flat prior on the initial state and allows the algorithm to learn entirely from the data [**durbin_koopman2012**]. In practice, $\kappa = 10^6$ or $\kappa = 10^7$ is used.

Signal-to-Noise Ratio (SNR)

The ratio between the variance of the state process and the variance of the measurement noise (the *signal-to-noise ratio*) is the parameter governing the filter's responsiveness. For the dynamic hedge ratio model with $\mathbf{T} = \mathbf{I}$:

$$\mathrm{SNR} = \frac{q_\beta}{\sigma_\varepsilon^2} \tag{19}$$

A high SNR implies that variations in the hedge ratio are large compared to measurement noise: the filter trusts recent observations more and rapidly updates its estimate of β_t (high responsiveness). A low SNR implies that observations are very noisy relative to parameter variations, so the filter relies more on the forecast and updates slowly (low responsiveness, smoother estimates). The optimal choice of SNR is an estimation problem solved by maximizing the log-likelihood of the state-space model with respect to $\boldsymbol{\psi} = (\sigma_\varepsilon^2, q_\alpha, q_\beta)$.

Pairs Trading Application

In the context of bivariate pairs trading, the state-space model is utilized as follows. The observed variable is the price of the first asset $y_t = Y_{1,t}$. The latent state vector $\boldsymbol{\alpha}_t = (\alpha_t, \beta_t)'$ contains the time-varying intercept and hedge ratio. The design matrix is the vector of regressors at time t: $\mathbf{Z}_t = (1, Y_{2,t})$. The model equations are:

$$Y_{1,t} = \begin{bmatrix} 1 & Y_{2,t} \end{bmatrix} \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{20}$$

$$\begin{bmatrix} \alpha_{t+1} \\ \beta_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} + \begin{bmatrix} \eta_{\alpha,t} \\ \eta_{\beta,t} \end{bmatrix}, \qquad \begin{bmatrix} \eta_{\alpha,t} \\ \eta_{\beta,t} \end{bmatrix} \sim \mathcal{N}(\mathbf{0},\, \mathbf{Q}) \tag{21}$$

with $\mathbf{Q} = \mathrm{diag}(q_\alpha, q_\beta)$ and $\mathbf{T} = \mathbf{I}_2$. The choice $\mathbf{T} = \mathbf{I}$ specifies that both α_t and β_t follow random walks: the parameter varies over time without mean reversion towards a fixed value, reflecting the lack of a priori reasons why the hedge ratio should return to a preset level.

4 Hidden Variables: Continuous vs. Discrete State Spaces

State-space models differ depending on the latent state space $\mathcal{X}$. The distinction between continuous and discrete state spaces reflects different assumptions about the DGP structure and requires different inference tools.

Continuous Spaces: Kalman Filter

When $\boldsymbol{\alpha}_t \in \mathbb{R}^m$, the latent state vector varies continuously over time. State trajectories are smooth in the sense that $\boldsymbol{\alpha}_{t+1}$ is close to $\boldsymbol{\alpha}_t$, modeling gradual parameter variations. The Kalman Filter provides the exact solution for this case (under Gaussianity): it calculates for each t the Gaussian conditional distribution of $\boldsymbol{\alpha}_t$ given $\boldsymbol{y}_{1:t}$, which is completely characterized by its filtered mean $\boldsymbol{a}_{t|t}$ and its filtered covariance $\mathbf{P}_{t|t}$.
Examples in the context of statistical arbitrage:

- **Dynamic hedge ratio:** β_t varies continuously in response to gradual shifts in the correlation structure between the two assets, modeled as a continuous random walk.
- **Stochastic volatility:** $\log \sigma_t^2$ follows a continuous AR(1) process, a non-parametric alternative to GARCH modeling of volatility.
- **Dynamic factor loadings:** the weights of a cointegration portfolio across k assets evolve as a continuous vector in $\mathbb{R}^k$, requiring the multivariate Kalman Filter.
- **Moving average and local level:** the long-term spread level varies slowly over time, identified by the local trend component of the state-space model.

Discrete Spaces: Hidden Markov Models

When $s_t \in \mathcal{S} = \{1, 2, \ldots, K\}$, the latent state process is a discrete variable identifying the current regime of the system. Shifts between regimes are discrete events, modeling sudden structural changes in the statistical properties of observed variables. Hidden Markov Models describe this

case. The state process is a hidden Markov chain that generates observations via regime-dependent emission distributions.
Examples in the context of statistical arbitrage:

- **Market regime:** $s_t = 1$ (low-volatility, stable cointegration, mean-reverting spread) or $s_t = 2$ (high-volatility, degraded cointegration, diverging spread). This is the most relevant case for pairs trading: operating in the wrong regime exposes the strategy to losses.
- **Business cycle:** $s_t \in \{1, 2, 3\}$ corresponding to expansion, contraction, and crisis. The statistical properties of sectoral returns and intersectoral cointegrating relationships vary across these states.
- **Market liquidity:** $s_t = 1$ (liquid market, tight bid-ask spread, negligible transaction costs) or $s_t = 2$ (illiquid market, wide bid-ask spread, transaction costs that could erode strategy profits).

The Hybrid Framework

The proposed framework combines both types of state spaces in a hierarchical two-level architecture. At the top level, the discrete regime s_t (governed by the HMM) determines the structure of emission distributions and state-dependent trading rules. At the lower level, continuous parameters $\boldsymbol{\alpha}_t = (\alpha_t, \beta_t)'$ (managed by the Kalman Filter) evolve within the current regime. This scheme allows capturing simultaneously the slow, gradual variation of the hedge ratio (continuous state space) and the discrete structural changes in the market regime (discrete state space).

5 The Kalman Filter

The Kalman Filter (KF), originally developed by Kalman for engineering applications [**kalman1960**] and adopted in econometrics by Harvey [**harvey1989**] and Durbin and Koopman [**durbin_koopman2012**], is a recursive algorithm that calculates, for each t, the conditional distribution of the state vector $\boldsymbol{\alpha}_t$ given the set of available observations $\boldsymbol{y}_{1:t}$. For the linear Gaussian model defined by equations (15)–(16), the KF provides the exact solution to the filtering problem: the conditional distribution of $\boldsymbol{\alpha}_t$ given $\boldsymbol{y}_{1:t}$ is Gaussian, and the KF recursively calculates its mean and covariance. No other linear estimator has a lower variance than the KF.
The algorithm operates in two alternating phases for each new observation. The *prediction step* projects the state estimate one step forward in time before observing $\boldsymbol{y}_t$, while the *update step* corrects the forecast by incorporating the information contained in the new observation.

From now on, we will use the convention from Durbin and Koopman [**durbin_koopman2012**]:

- $\boldsymbol{a}_{t|t-1} = \mathbb{E}[\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t-1}]$: a priori state estimate at time t, based on information up to $t-1$.
- $\mathbf{P}_{t|t-1} = \mathrm{Var}(\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t-1})$: a priori covariance matrix (uncertainty on the estimate before observing $\boldsymbol{y}_t$).
- $\boldsymbol{a}_{t|t} = \mathbb{E}[\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t}]$: a posteriori (filtered) state estimate after incorporating $\boldsymbol{y}_t$.
- $\mathbf{P}_{t|t} = \mathrm{Var}(\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t})$: a posteriori covariance matrix (residual uncertainty after the update).
- $\boldsymbol{v}_t = \boldsymbol{y}_t - \mathbf{Z}_t \boldsymbol{a}_{t|t-1} - \mathbf{d}_t$: innovation, i.e., the one-step-ahead forecast error.
- $\mathbf{F}_t = \mathrm{Var}(\boldsymbol{v}_t \mid \boldsymbol{y}_{1:t-1}) = \mathbf{Z}_t \mathbf{P}_{t|t-1} \mathbf{Z}_t' + \mathbf{H}_t$: innovation covariance matrix.
- $\mathbf{K}_t = \mathbf{P}_{t|t-1} \mathbf{Z}_t' \mathbf{F}_t^{-1}$: Kalman gain, an $m \times p$ matrix that determines the weight of new observations in updating the estimate.

Prediction Step

Starting from the filtered estimate $\boldsymbol{a}_{t-1|t-1}$ and covariance $\mathbf{P}_{t-1|t-1}$ at time $t-1$, the prediction step projects the state at time t using the transition equation:

$$\boldsymbol{a}_{t|t-1} = \mathbf{T}_{t-1}\,\boldsymbol{a}_{t-1|t-1} + \mathbf{c}_{t-1} \tag{22}$$

$$\mathbf{P}_{t|t-1} = \mathbf{T}_{t-1}\,\mathbf{P}_{t-1|t-1}\,\mathbf{T}'_{t-1} + \mathbf{R}_{t-1}\,\mathbf{Q}_{t-1}\,\mathbf{R}'_{t-1} \tag{23}$$

Equation (22) applies the linear transformation $\mathbf{T}_{t-1}$ to the previous estimate: it is the optimal forecast of the future state given all past information. Equation (23) shows how uncertainty propagates over time: the *a priori* covariance is the linear transformation of the previous *a posteriori* covariance, augmented by the contribution of process noise $\mathbf{R}_{t-1}\mathbf{Q}_{t-1}\mathbf{R}'_{t-1}$. This latter term is crucial: without process noise ($\mathbf{Q} = \mathbf{0}$), the state would be deterministic and would never change; process noise is the mechanism by which the filter remains open to updating even after many observations.

Update Step

When the observation $\boldsymbol{y}_t$ becomes available, the filter updates the state estimate by combining the prior prediction with the information contained in the new observation:

$$\boldsymbol{v}_t = \boldsymbol{y}_t - \mathbf{Z}_t\,\boldsymbol{a}_{t|t-1} - \mathbf{d}_t \tag{24}$$

$$\mathbf{F}_t = \mathbf{Z}_t\,\mathbf{P}_{t|t-1}\,\mathbf{Z}'_t + \mathbf{H}_t \tag{25}$$

$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\,\mathbf{Z}'_t\,\mathbf{F}_t^{-1} \tag{26}$$

$$\boldsymbol{a}_{t|t} = \boldsymbol{a}_{t|t-1} + \mathbf{K}_t\,\boldsymbol{v}_t \tag{27}$$

$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\,\mathbf{Z}_t)\,\mathbf{P}_{t|t-1} \tag{28}$$

Kalman Gain

The Kalman gain $\mathbf{K}_t$ determines the relative weight assigned to the prior prediction versus the new observation. Rewriting equation (27):

$$\boldsymbol{a}_{t|t} = \underbrace{(\mathbf{I} - \mathbf{K}_t\mathbf{Z}_t)\,\boldsymbol{a}_{t|t-1}}_{\text{weight on prediction}} + \underbrace{\mathbf{K}_t\,\boldsymbol{y}_t}_{\text{weight on observation}} \tag{29}$$

The a posteriori estimate is a weighted average of the prediction and the new observation (with weights determined by the Kalman gain). If measurement noise $\mathbf{H}_t$ is very large compared to uncertainty $\mathbf{P}_{t|t-1}$ — observations are highly noisy — then $\mathbf{F}_t \approx \mathbf{H}_t$ is large, $\mathbf{K}_t \approx \mathbf{0}$, and the filter practically ignores the new observation, preferring the previous prediction. Conversely, if $\mathbf{H}_t \to \mathbf{0}$ — observations are practically noise-free — then $\mathbf{K}_t \to (\mathbf{Z}'_t\mathbf{Z}_t)^{-1}\mathbf{Z}'_t$ and the filter fully trusts the current observation. The Kalman gain balances these two sources of information.

Posterior Covariance and Noise Reduction

Equation (28) shows that the a posteriori covariance matrix is always smaller (in a matrix sense) than the a priori covariance: $\mathbf{P}_{t|t} \preceq \mathbf{P}_{t|t-1}$. Every new observation reduces uncertainty regarding the state estimate. The variance reduction is given by:

$$\mathbf{P}_{t|t-1} - \mathbf{P}_{t|t} = \mathbf{K}_t\,\mathbf{F}_t\,\mathbf{K}'_t = \mathbf{P}_{t|t-1}\,\mathbf{Z}'_t\,\mathbf{F}_t^{-1}\,\mathbf{Z}_t\,\mathbf{P}_{t|t-1} \tag{30}$$

However, the prediction step (23) increases covariance by adding process noise $\mathbf{RQR}'$. In a stationary regime, the alternation of prediction (uncertainty increase) and update (uncertainty reduction)

leads to an asymptotically constant state covariance (known as steady-state covariance) that satisfies the Riccati equation:

$$\mathbf{P}_\infty = \mathbf{T}\left(\mathbf{P}_\infty - \mathbf{P}_\infty \mathbf{Z}' \mathbf{F}_\infty^{-1} \mathbf{Z} \mathbf{P}_\infty\right) \mathbf{T}' + \mathbf{R}\mathbf{Q}\mathbf{R}' \tag{31}$$

Under this condition, the Kalman gain stabilizes at $\mathbf{K}_\infty = \mathbf{P}_\infty \mathbf{Z}' \mathbf{F}_\infty^{-1}$ and no longer varies over time.

Log-Likelihood and Parameter Estimation

The innovations $\boldsymbol{v}_t$ produced by the KF are serially uncorrelated. $\mathbb{E}[\boldsymbol{v}_t \boldsymbol{v}_s'] = \mathbf{0}$ for $t \neq s$. This property is the orthogonality of innovations, which is a direct consequence of orthogonal projection and allows the joint density of observations to be factorized:

$$p(\boldsymbol{y}_{1:T} \mid \boldsymbol{\psi}) = \prod_{t=1}^{T} p(\boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1}, \boldsymbol{\psi}) = \prod_{t=1}^{T} \mathcal{N}(\boldsymbol{y}_t;\, \mathbf{Z}_t \boldsymbol{a}_{t|t-1} + \mathbf{d}_t,\, \mathbf{F}_t) \tag{32}$$

The log-likelihood is obtained using the prediction error decomposition [**harvey1989**]:

$$\log \mathcal{L}(\boldsymbol{\psi}) = -\frac{Tp}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^{T} \left[\log |\mathbf{F}_t(\boldsymbol{\psi})| + \boldsymbol{v}_t(\boldsymbol{\psi})' \, \mathbf{F}_t(\boldsymbol{\psi})^{-1} \, \boldsymbol{v}_t(\boldsymbol{\psi})\right] \tag{33}$$

where the dependence of $\boldsymbol{v}_t$ and $\mathbf{F}_t$ on the parameters $\boldsymbol{\psi} = (\sigma_\varepsilon^2, q_\alpha, q_\beta)$ is explicitly shown. Maximizing (33) with respect to $\boldsymbol{\psi}$ (typically using gradient-based algorithms like L-BFGS-B with multiple starting points to avoid local maxima) yields the MLE estimates of the model parameters.

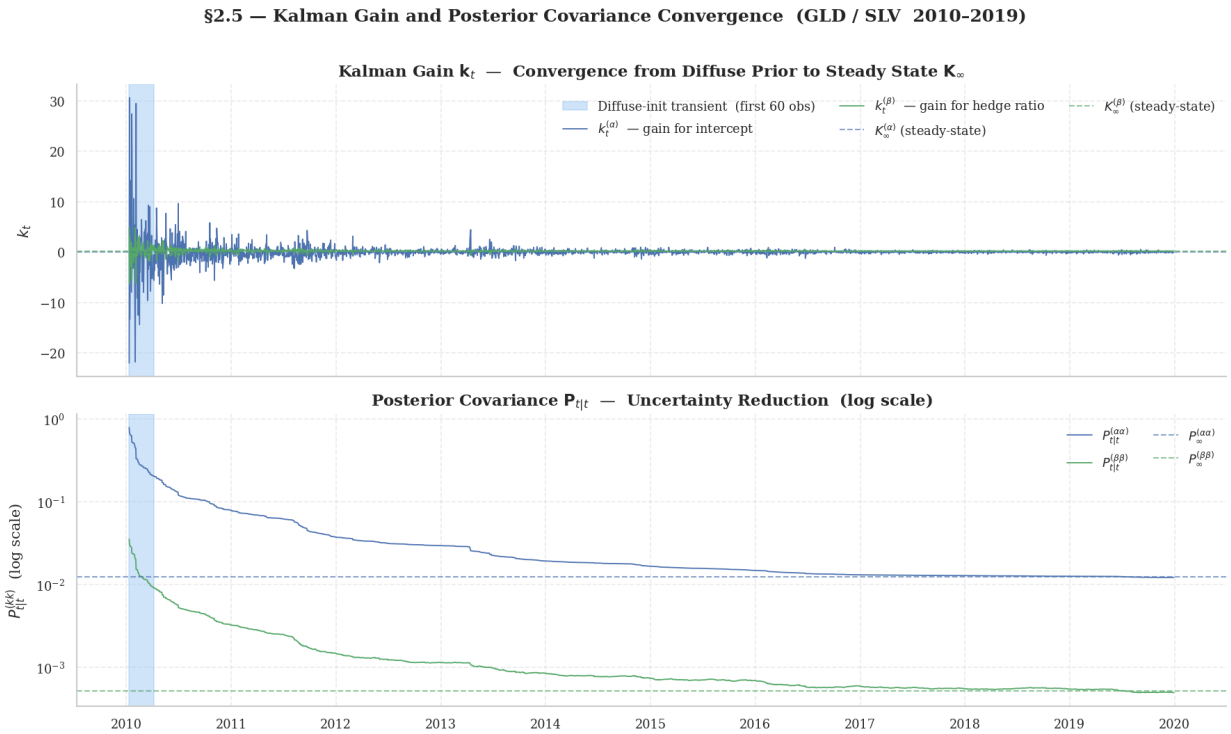


Figure 2: Top panel: Kalman gain components $\mathbf{k}_t$ for the intercept ($k_t^{(\alpha)}$) and the hedge ratio ($k_t^{(\beta)}$) on GLD/SLV (2010–2019). The blue area highlights the initial transient due to diffuse initialization; dashed lines indicate steady-state values $K_\infty^{(\alpha)}$ and $K_\infty^{(\beta)}$. Bottom panel: *a posteriori* covariance $\mathbf{P}_{t|t}$ on a logarithmic scale. The monotonic convergence towards P_∞ (dashed lines) represents the empirical solution of the Riccati equation and certifies that the filter has reached a steady state.

Table 1: MLE parameters of the state-space model and estimated steady-state values on GLD/SLV (2010–2019), $T = 2{,}515$ daily observations. Parameters $\hat{\sigma}_\varepsilon^2$, $\hat{q}_\alpha$, $\hat{q}_\beta$ are estimated by maximizing the log-likelihood using L-BFGS-B with 5 starting points. Steady-state values K_∞ and P_∞ are calculated as the average over the last 10% of the sample.

Parameter	**Description**	**Estimated Value**
MLE Parameters		
$\hat{\sigma}_\varepsilon^2$	Measurement noise variance	5.995×10^{-9}
$\hat{q}_\alpha$	State process variance (intercept)	1.650×10^{-6}
$\hat{q}_\beta$	State process variance (hedge ratio)	4.391×10^{-6}
SNR	Signal-to-noise ratio $\hat{q}_\beta / \hat{\sigma}_\varepsilon^2$	732.5
Steady-state Kalman Gain K_∞		
$K_\infty^{(\alpha)}$	Steady-state gain for intercept	2.135×10^{-4}
$K_\infty^{(\beta)}$	Steady-state gain for hedge ratio	2.050×10^{-1}
Steady-state Posterior Covariance P_∞		
$P_\infty^{(\alpha\alpha)}$	Steady-state covariance for intercept	1.234×10^{-2}
$P_\infty^{(\beta\beta)}$	Steady-state covariance for hedge ratio	5.194×10^{-4}
Dynamic Spread $S_t = v_t$		
$\min(S_t)$	Minimum value of KF spread	-0.0781
$\max(S_t)$	Maximum value of KF spread	$+0.0459$
$\min(z_t^{\mathrm{KF}})$	Minimum value of KF z-score	-7.38
$\max(z_t^{\mathrm{KF}})$	Maximum value of KF z-score	$+4.34$

Kalman Smoother

The KF produces filtered estimates $\boldsymbol{a}_{t|t}$ leveraging only observations up to time t. For historical analysis and model calibration, it is useful to compute smoothed estimates $\boldsymbol{a}_{t|T}$ that leverage the entire time series $\boldsymbol{y}_{1:T}$ to infer the state at time t. The Kalman smoother by Rauch, Tung, and Striebel [**rauch1965**] computes smoothed estimates via a backward recursion starting from $t = T$ down to $t = 1$:

$$\mathbf{L}_t = \mathbf{P}_{t|t}\, \mathbf{T}_t'\, \mathbf{P}_{t+1|t}^{-1} \tag{34}$$

$$\boldsymbol{a}_{t|T} = \boldsymbol{a}_{t|t} + \mathbf{L}_t \left(\boldsymbol{a}_{t+1|T} - \boldsymbol{a}_{t+1|t} \right) \tag{35}$$

$$\mathbf{P}_{t|T} = \mathbf{P}_{t|t} + \mathbf{L}_t \left(\mathbf{P}_{t+1|T} - \mathbf{P}_{t+1|t} \right) \mathbf{L}_t' \tag{36}$$

Smoothed estimates $\boldsymbol{a}_{t|T}$ cannot be used for real-time trading (as they require future observations) but are fundamental for ex-post analysis and model optimization.

6 Recursive Bayesian Estimation

The Kalman Filter can be derived within the Bayesian framework as a special case of the recursive update of the posterior state distribution.

The filtering problem consists of calculating the posterior distribution of the latent state at time t given all information up to t:

$$p(\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t}) \propto p(\boldsymbol{y}_t \mid \boldsymbol{\alpha}_t)\, p(\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t-1}) \tag{37}$$

Equation (37) is simply Bayes' theorem applied to the filtering problem. The posterior distribution is proportional to the product of the likelihood of the current observation and the prior predictive

distribution. The prior predictive distribution is obtained by integrating out the previous state:

$$p(\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t-1}) = \int p(\boldsymbol{\alpha}_t \mid \boldsymbol{\alpha}_{t-1})\, p(\boldsymbol{\alpha}_{t-1} \mid \boldsymbol{y}_{1:t-1})\, d\boldsymbol{\alpha}_{t-1} \tag{38}$$

Equations (37)–(38) define the recursive Bayesian filter. In the absence of Gaussianity and linearity assumptions, these equations lack closed-form analytical solutions and require approximation methods such as the Particle Filter or Unscented Kalman Filter. Under Gaussianity and linearity, however, the analytical solution exists and is the Kalman Filter.

Assume that the filtered distribution at time $t-1$ is Gaussian:

$$p(\boldsymbol{\alpha}_{t-1} \mid \boldsymbol{y}_{1:t-1}) = \mathcal{N}(\boldsymbol{\alpha}_{t-1};\, \boldsymbol{a}_{t-1|t-1},\, \mathbf{P}_{t-1|t-1}) \tag{39}$$

Prediction Step: Since the transition $p(\boldsymbol{\alpha}_t \mid \boldsymbol{\alpha}_{t-1}) = \mathcal{N}(\boldsymbol{\alpha}_t;\, \mathbf{T}\boldsymbol{\alpha}_{t-1} + \mathbf{c},\, \mathbf{RQR}')$ is Gaussian and the convolution of Gaussians is Gaussian, the integral (38) can be computed analytically:

$$\begin{aligned} p(\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t-1}) &= \int \mathcal{N}(\mathbf{T}\boldsymbol{\alpha}_{t-1} + \mathbf{c},\, \mathbf{RQR}')\, \mathcal{N}(\boldsymbol{a}_{t-1|t-1},\, \mathbf{P}_{t-1|t-1})\, d\boldsymbol{\alpha}_{t-1} \\ &= \mathcal{N}\big(\boldsymbol{\alpha}_t;\, \mathbf{T}\boldsymbol{a}_{t-1|t-1} + \mathbf{c},\, \mathbf{T}\mathbf{P}_{t-1|t-1}\mathbf{T}' + \mathbf{RQR}'\big) \\ &= \mathcal{N}\big(\boldsymbol{\alpha}_t;\, \boldsymbol{a}_{t|t-1},\, \mathbf{P}_{t|t-1}\big) \end{aligned} \tag{40}$$

which exactly coincides with equations (22)–(23).
Update Step: The joint distribution of $(\boldsymbol{\alpha}_t, \boldsymbol{y}_t)$ given $\boldsymbol{y}_{1:t-1}$ is bivariate Gaussian. Using the properties of the joint Gaussian distribution, the covariance between $\boldsymbol{\alpha}_t$ and $\boldsymbol{y}_t$ is:

$$\mathrm{Cov}(\boldsymbol{\alpha}_t, \boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1}) = \mathrm{Cov}(\boldsymbol{\alpha}_t, \mathbf{Z}_t\boldsymbol{\alpha}_t + \boldsymbol{\varepsilon}_t) = \mathbf{P}_{t|t-1}\mathbf{Z}_t' \tag{41}$$

The conditional distribution of $\boldsymbol{\alpha}_t$ given $\boldsymbol{y}_t$ (and $\boldsymbol{y}_{1:t-1}$) is obtained by applying the formula for the conditional of a bivariate Gaussian:

$$\mathbb{E}[\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t}] = \boldsymbol{a}_{t|t-1} + \mathbf{P}_{t|t-1}\mathbf{Z}_t'\, \mathbf{F}_t^{-1}\, \boldsymbol{v}_t = \boldsymbol{a}_{t|t} \tag{42}$$

$$\mathrm{Var}(\boldsymbol{\alpha}_t \mid \boldsymbol{y}_{1:t}) = \mathbf{P}_{t|t-1} - \mathbf{P}_{t|t-1}\mathbf{Z}_t'\, \mathbf{F}_t^{-1}\, \mathbf{Z}_t\mathbf{P}_{t|t-1} = \mathbf{P}_{t|t} \tag{43}$$

which coincide with equations (27)–(28). The Gaussian functional form is preserved at each iteration. Starting from a Gaussian prior, the posterior is Gaussian, which then becomes the prior for the next step. This closure result (the conjugate update) is the fundamental property making the KF computationally tractable and exact.

Theorem 6.1 (Optimality of the Kalman Filter)**.** *Under the linear Gaussian state-space model* (15)–(16)*, the Kalman Filter is the optimal estimator of* $\boldsymbol{\alpha}_t$ *in the sense of Minimum Mean Square Error (MMSE) among all unbiased, linear, and non-linear estimators.*

Under non-Gaussianity, the KF remains the Best Linear Unbiased Estimator (BLUE) but there might exist a non-linear estimator with lower mean squared error.

7 Dynamic Hedge Ratio

The application of the Kalman Filter to pairs trading involves recursively estimating the hedge ratio β_t and intercept α_t. The idea is to treat the cointegrating regression as a linear regression model where coefficients vary over time following a random walk, applying the KF to estimate them optimally given the set of all observations up to t.

Model Specification

The state-space model for the dynamic hedge ratio is defined by the following equations. Let's simplify the notation by omitting the intercept and focusing solely on the hedge ratio:

$$Y_{1,t} = \mathbf{z}_t' \boldsymbol{\alpha}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{44}$$

$$\boldsymbol{\alpha}_{t+1} = \boldsymbol{\alpha}_t + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \tag{45}$$

with $\mathbf{z}_t = (1, Y_{2,t})'$, $\boldsymbol{\alpha}_t = (\alpha_t, \beta_t)'$, and $\mathbf{Q} = \text{diag}(q_\alpha, q_\beta)$. The parameters to be estimated via MLE are $\boldsymbol{\psi} = (\sigma_\varepsilon^2, q_\alpha, q_\beta)$.

KF Recursion on the Hedge Ratio

For the univariate case $p = 1$, KF equations simplify to scalar form. Let $\mathbf{z}_t = \mathbf{Z}_t'$ for consistency with vector notation:

Initialization: $\boldsymbol{a}_{1|0} = \mathbf{0}$, $\mathbf{P}_{1|0} = \kappa \mathbf{I}_2$ with $\kappa = 10^6$.

For $t = 1, \ldots, T$*:*

$$\begin{aligned}
v_t &= Y_{1,t} - \mathbf{z}_t' \boldsymbol{a}_{t|t-1} && \text{(Innovation)} \\
f_t &= \mathbf{z}_t' \mathbf{P}_{t|t-1} \mathbf{z}_t + \sigma_\varepsilon^2 && \text{(Innovation variance)} \\
\mathbf{k}_t &= \mathbf{P}_{t|t-1} \mathbf{z}_t / f_t && \text{(Kalman gain)} \\
\boldsymbol{a}_{t|t} &= \boldsymbol{a}_{t|t-1} + \mathbf{k}_t \, v_t && \text{(State update)} \\
\mathbf{P}_{t|t} &= \left(\mathbf{I}_2 - \mathbf{k}_t \mathbf{z}_t'\right) \mathbf{P}_{t|t-1} && \text{(Covariance update)} \\
\boldsymbol{a}_{t+1|t} &= \boldsymbol{a}_{t|t} && \text{(State prediction)} \\
\mathbf{P}_{t+1|t} &= \mathbf{P}_{t|t} + \mathbf{Q} && \text{(Covariance prediction)}
\end{aligned}$$

Dynamic Spread and Adaptive Z-score

The filtered estimate $\hat{\beta}_{t|t} = [\boldsymbol{a}_{t|t}]_2$ (second component of the filtered state vector) allows us to define the dynamic spread:

$$S_t = Y_{1,t} - \hat{\alpha}_{t|t} - \hat{\beta}_{t|t} \, Y_{2,t} = v_t \tag{46}$$

The dynamic spread coincides with the filter's innovation v_t. It is indeed the component of $Y_{1,t}$ unexplained by the local regression updated to time t. This coincidence is a consequence of the one-step-ahead forecast structure in the state-space model. The conditional variance of the spread is f_t, and the adaptive z-score is:

$$z_t = \frac{v_t}{\sqrt{f_t}} = \frac{S_t}{\sqrt{f_t}} \tag{47}$$

Unlike the static z-score, which normalizes using a constant sample standard deviation $\hat{\sigma}_\varepsilon$, the adaptive z-score (47) employs $\sqrt{f_t}$, which varies over time depending on current uncertainty. During periods when $\mathbf{P}_{t|t-1}$ is large (high parametric uncertainty), f_t grows larger, thereby shrinking the z-score and reducing the probability of false signals. This self-calibrating property differentiates the KF approach from classical strategies.

Adaptive Confidence Bands

The innovation variance f_t provides a confidence band for the spread:

$$\hat{Y}_{1,t} \pm z_{\alpha/2} \sqrt{f_t} \tag{48}$$

where $\hat{Y}_{1,t} = \mathbf{z}_t' \boldsymbol{a}_{t|t-1}$ is the one-step-ahead forecast of $Y_{1,t}$. This band provides exact $(1 - \alpha)$ coverage under the Gaussian state-space model. It widens during periods of elevated parametric

uncertainty and narrows when uncertainty is low, furnishing a risk metric that automatically adapts to market conditions.

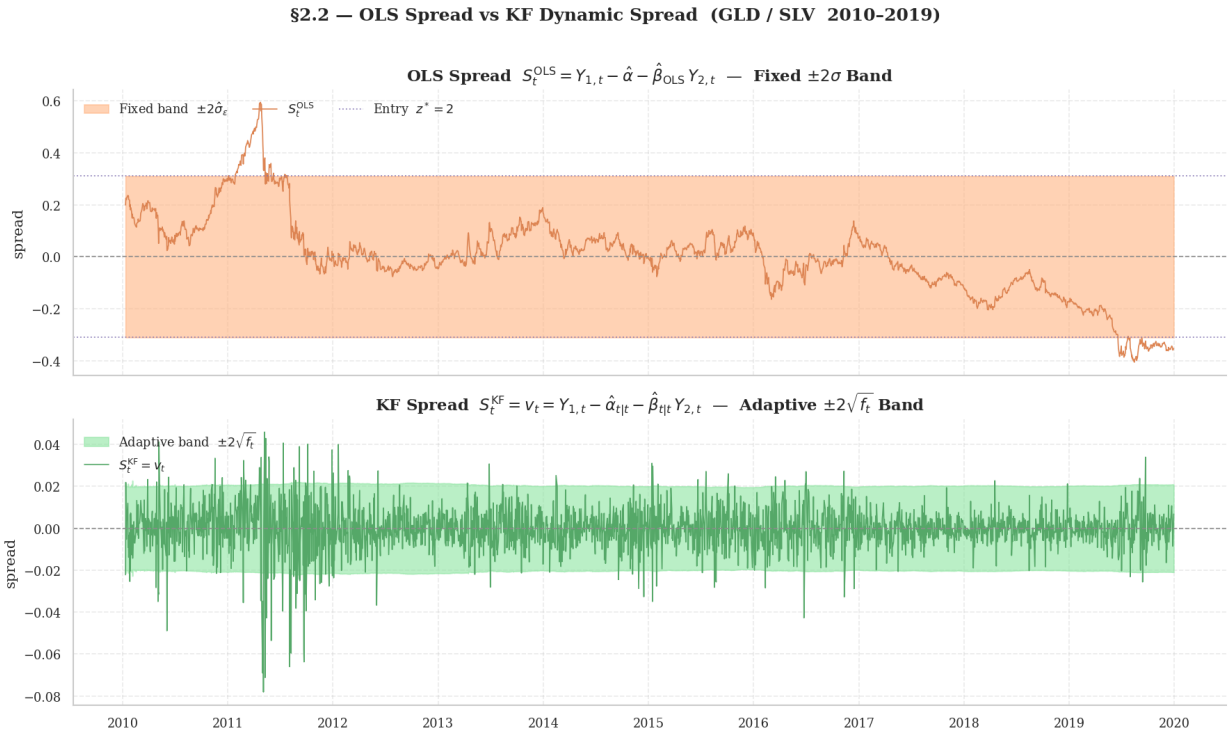


Figure 3: Comparison between static z-score $z_t^{\text{OLS}} = S_t^{\text{OLS}} / \hat{\sigma}_\varepsilon$ (top panel) and adaptive z-score $z_t^{\text{KF}} = v_t / \sqrt{f_t}$ (bottom panel) on GLD/SLV (2010–2019). The pink area highlights the high volatility episode from August 2015 to February 2016. The static z-score exhibits a persistent negative drift in the second half of the sample caused by incorrect hedge ratio estimation, generating false short entry signals. The adaptive z-score remains stationary around zero throughout the sample, with the conditional standard deviation $\sqrt{f_t}$ self-calibrating to the current market context.

8 Markov Chains and Hidden Markov Models

While the Kalman Filter governs the continuous evolution of parameters, Hidden Markov Models (HMM) model the (discrete) latent state variable $s_t \in \mathcal{S} = \{1, 2, \ldots, K\}$ identifying the current market regime. HMMs were introduced in signal analysis by Baum and Petrie [**baum1966**, **baum1970**], later extended by Rabiner [**rabiner1989**], and applied to finance by Ryden, Teräsvirta, and Åsbrink [**ryden1998**] and Ang and Bekaert [**ang_bekaert2002**].

Properties of Markov Chains

A discrete Markov chain over $\mathcal{S} = \{1, \ldots, K\}$ is entirely characterized by two objects. The first is the initial distribution $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)'$ with $\pi_k = \Pr(s_1 = k)$ and $\sum_k \pi_k = 1$. The second is the transition matrix $\mathbf{P} \in [0,1]^{K \times K}$ with elements:

$$p_{ij} = \Pr(s_{t+1} = j \mid s_t = i), \quad \sum_{j=1}^{K} p_{ij} = 1 \quad \forall i \tag{49}$$

Row i of $\mathbf{P}$ is the probability distribution of states at time $t+1$ when the current state is i. The diagonal entry p_{ii} measures the persistence of regime i: the expected duration in regime i prior to a transition is geometrically distributed with mean $d_i = 1/(1 - p_{ii})$. For typical values in finance, $p_{11} \approx 0.97$ for the low-volatility regime (average duration ≈ 33 days) and $p_{22} \approx 0.90$ for

the high-volatility regime (average duration ≈ 10 days). This reflects the asymmetry documented empirically: markets spend prolonged periods in low volatility interspersed with shorter but more intense stress episodes.
The stationary distribution $\boldsymbol{\pi}^*$ satisfies $\boldsymbol{\pi}^* = \boldsymbol{\pi}^*\mathbf{P}$ and is the left eigenvector of $\mathbf{P}$ associated with the unit eigenvalue. For a bivariate HMM ($K = 2$) with transition matrix:

$$\mathbf{P} = \begin{bmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{bmatrix} \tag{50}$$

the stationary distribution is given by:

$$\pi_1^* = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}, \qquad \pi_2^* = \frac{1 - p_{11}}{2 - p_{11} - p_{22}} \tag{51}$$

representing the expected proportion of time spent in each regime over the long term.

Definition of HMM

A Hidden Markov Model is a Markov chain with latent (unobservable) states coupled to an emission process that generates observations. An HMM is defined by a triplet $\boldsymbol{\lambda} = (\mathbf{P}, \boldsymbol{\pi}, \{f_k\}_{k=1}^K)$ where:

- $\mathbf{P}$ is the latent state transition matrix.
- $\boldsymbol{\pi}$ is the initial state distribution.
- $f_k(\boldsymbol{y}_t) = p(\boldsymbol{y}_t \mid s_t = k; \boldsymbol{\theta}_k)$ is the emission density of regime k, parameterized by $\boldsymbol{\theta}_k$.

Observations $\boldsymbol{y}_t$ are conditionally independent given the sequence of states $p(\boldsymbol{y}_t \mid s_t, s_{t-1}, \ldots, s_1, \boldsymbol{y}_{1:t-1}) = p(\boldsymbol{y}_t \mid s_t)$. Usually, regime-dependent Gaussian emission is assumed:

$$\boldsymbol{y}_t \mid s_t = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{52}$$

The unconditional distribution of observations is a Gaussian mixture:

$$p(\boldsymbol{y}_t) = \sum_{k=1}^{K} \pi_k^* \mathcal{N}(\boldsymbol{y}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{53}$$

which provides a formal explanation for the non-Gaussianity documented in financial returns: unconditional excess kurtosis emerges from the mixture structure, even when each component is Gaussian.

3 Main Issues with HMMs

The practical use of HMMs requires solving three computational problems identified by Rabiner [**rabiner1989**]:
The **first problem** (Evaluation): given the model $\boldsymbol{\lambda}$ and an observation sequence $\boldsymbol{y}_{1:T}$, what is the probability $p(\boldsymbol{y}_{1:T} \mid \boldsymbol{\lambda})$? This is used to compare models with different numbers of regimes and for MLE parameter estimation.
The **second problem** (Decoding): given an observation sequence $\boldsymbol{y}_{1:T}$, what is the most likely latent state sequence $\hat{s}_{1:T}^* = \arg\max_{s_{1:T}} p(s_{1:T} \mid \boldsymbol{y}_{1:T}, \boldsymbol{\lambda})$? The answer provides a segmentation of the time series into market regimes.
The **third problem** (Learning): how to estimate model parameters $\boldsymbol{\lambda}$ from an observation sequence $\boldsymbol{y}_{1:T}$? This is the estimation problem, solved by the Baum-Welch algorithm.

Forward Algorithm: Solution for the First Problem

The solution to the first problem relies on defining the forward variable:

$$\alpha_t(k) = p(\boldsymbol{y}_{1:t},\, s_t = k \mid \boldsymbol{\lambda}) \tag{54}$$

which represents the joint probability of observing the sequence $\boldsymbol{y}_{1:t}$ and being in state k at time t.
Initialization:

$$\alpha_1(k) = \pi_k\, f_k(\boldsymbol{y}_1), \qquad k = 1, \ldots, K \tag{55}$$

Recursion ($t = 2, \ldots, T$):

$$\alpha_t(j) = \left[\sum_{i=1}^{K} \alpha_{t-1}(i)\, p_{ij}\right] f_j(\boldsymbol{y}_t), \qquad j = 1, \ldots, K \tag{56}$$

Termination:

$$p(\boldsymbol{y}_{1:T} \mid \boldsymbol{\lambda}) = \sum_{k=1}^{K} \alpha_T(k) \tag{57}$$

Recursion (56) sums, for each state j at time t, the probability of having been in any state i at time $t-1$ (weighted by transition probability p_{ij}) and multiplies by the emission probability $f_j(\boldsymbol{y}_t)$. The computational cost is $O(K^2T)$, compared to the $O(K^T)$ complexity of enumerating all state sequences.

Backward Algorithm

The backward variable is defined as:

$$\beta_t(k) = p(\boldsymbol{y}_{t+1:T} \mid s_t = k,\, \boldsymbol{\lambda}) \tag{58}$$

representing the probability of observing the future part of the sequence $\boldsymbol{y}_{t+1:T}$ given that at time t the system is in state k.
Initialization: $\beta_T(k) = 1$ for all k.
Recursion ($t = T-1, \ldots, 1$):

$$\beta_t(i) = \sum_{j=1}^{K} p_{ij}\, f_j(\boldsymbol{y}_{t+1})\, \beta_{t+1}(j) \tag{59}$$

The backward variable mirrors the forward variable: it sums, for each state j at time $t+1$, the probability of transitioning from i to j (weighted by j's emission at $t+1$) multiplied by the probability of continuing the path from j onwards.
Forward and backward quantities combine to produce smoothed regime probabilities:

$$\gamma_t(k) = \Pr(s_t = k \mid \boldsymbol{y}_{1:T},\, \boldsymbol{\lambda}) = \frac{\alpha_t(k)\, \beta_t(k)}{\sum_{j=1}^{K} \alpha_t(j)\, \beta_t(j)} \tag{60}$$

and joint transition probabilities:

$$\xi_t(i,j) = \Pr(s_t = i,\, s_{t+1} = j \mid \boldsymbol{y}_{1:T},\, \boldsymbol{\lambda}) = \frac{\alpha_t(i)\, p_{ij}\, f_j(\boldsymbol{y}_{t+1})\, \beta_{t+1}(j)}{p(\boldsymbol{y}_{1:T} \mid \boldsymbol{\lambda})} \tag{61}$$

Viterbi Algorithm: Solution for the Second Problem

The Viterbi algorithm [**viterbi1967**] solves the second problem by finding the most likely sequence of states using dynamic programming. Let's define:

$$\delta_t(k) = \max_{s_{1:t-1}} p(\boldsymbol{y}_{1:t},\, s_{1:t-1},\, s_t = k \mid \boldsymbol{\lambda}) \tag{62}$$

as the probability of the most likely path ending in state k at time t:
Initialization: $\delta_1(k) = \pi_k\, f_k(\boldsymbol{y}_1)$, $\psi_1(k) = 0$.
Recursion ($t = 2, \ldots, T$)*:*

$$\delta_t(j) = \max_{i \in \mathcal{S}} \left[\delta_{t-1}(i)\, p_{ij}\right] f_j(\boldsymbol{y}_t) \tag{63}$$

$$\psi_t(j) = \arg\max_{i \in \mathcal{S}} \left[\delta_{t-1}(i)\, p_{ij}\right] \tag{64}$$

Termination and Backtracking:

$$\hat{s}_T^* = \arg\max_k \delta_T(k), \qquad \hat{s}_t^* = \psi_{t+1}(\hat{s}_{t+1}^*) \quad \text{for } t = T-1, \ldots, 1 \tag{65}$$

Pointers $\psi_t(j)$ store the optimal state at the previous step: backtracking reconstructs the most probable sequence backwards. Note that the Viterbi sequence $\hat{s}_{1:T}^*$ is globally optimal (maximizes the joint probability $p(s_{1:T} \mid \boldsymbol{y}_{1:T})$) but can differ from the sequence obtained by selecting, at each t, the regime with maximal marginal probability (which maximizes $\sum_t \Pr(s_t = k \mid \boldsymbol{y}_{1:T})$ but guarantees no sequence coherence).

Baum-Welch Algorithm: Solution for the Third Problem

The Baum-Welch algorithm is a special case of the Expectation-Maximization (EM) algorithm [**dempster1977**] for MLE estimation of HMM parameters. It alternates between an E step (computing expectations of latent quantities given current models) and an M step (maximizing the complete log-likelihood with respect to parameters).
E Step: Using current parameters $\boldsymbol{\lambda}^{(s)}$, forward (56) and backward (59) recursions are run, and $\gamma_t(k)$ (60) and $\xi_t(i,j)$ (61) are calculated for each t and pair (i,j).
M Step: Parameters are updated to maximize the expected complete log-likelihood. Closed-form update formulas are:

$$\hat{p}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \tag{66}$$

$$\hat{\boldsymbol{\mu}}_k = \frac{\sum_{t=1}^{T} \gamma_t(k)\, \boldsymbol{y}_t}{\sum_{t=1}^{T} \gamma_t(k)} \tag{67}$$

$$\hat{\boldsymbol{\Sigma}}_k = \frac{\sum_{t=1}^{T} \gamma_t(k)\; (\boldsymbol{y}_t - \hat{\boldsymbol{\mu}}_k)\, (\boldsymbol{y}_t - \hat{\boldsymbol{\mu}}_k)'}{\sum_{t=1}^{T} \gamma_t(k)} \tag{68}$$

The numerator in (66) is the expected number of transitions from regime i to j; the denominator is the expected number of periods spent in regime i. The ratio estimates the transition probability. Similarly, $\hat{\boldsymbol{\mu}}_k$ is the weighted average of observations with weight $\gamma_t(k)$ equal to the posterior probability of being in regime k at time t.
The algorithm guarantees the non-decrease of the log-likelihood at each iteration and converges to a local maximum (not necessarily global). Sensitivity to initial conditions is its limitation.

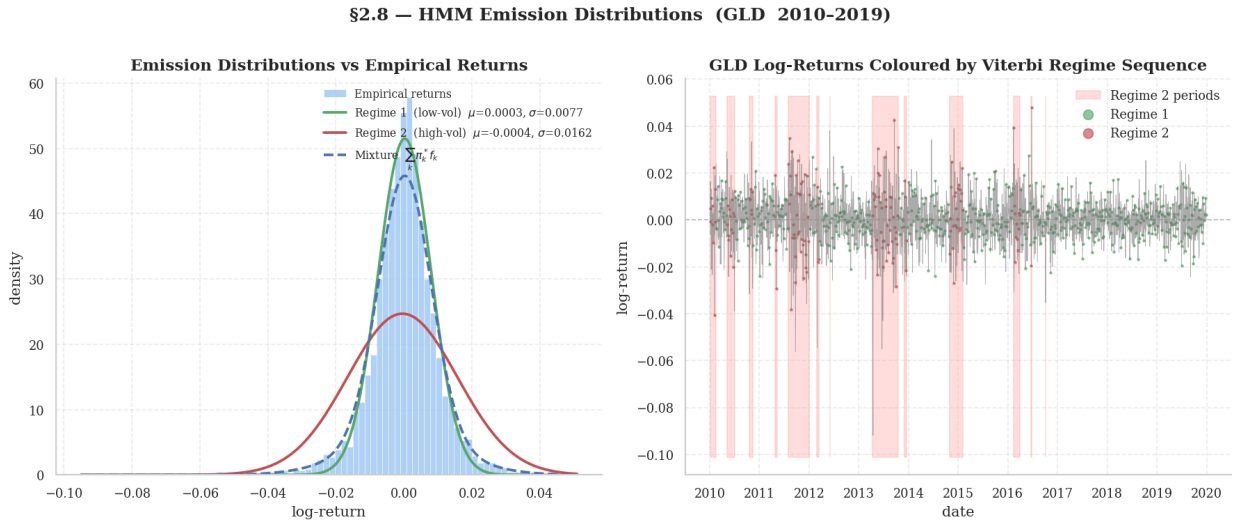


Figure 4: Left: regime-dependent emission densities $f_k(\boldsymbol{y}_t) = \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ estimated via Baum-Welch on GLD (2010–2019), with the unconditional mixture distribution $\sum_k \pi_k^* f_k$ overlaid on the empirical returns histogram. Right: GLD log-returns time series colored by regime according to the Viterbi sequence $\hat{s}_{1:T}^*$; shaded areas indicate periods classified as regime 2 (high volatility).

Table 2: Estimated parameters for the two-state HMM on GLD log-returns (2010–2019), $T = 2{,}514$ daily returns. Estimation is conducted via the Baum-Welch algorithm with 50 random initializations; the maximum log-likelihood solution is reported. Expected duration $d_k = 1/(1-p_{kk})$ is expressed in trading days.

Parameter	**Regime 1 (Low Vol)**	**Regime 2 (High Vol)**
$\hat{\mu}_k$	$+0.00025$	-0.00044
$\hat{\sigma}_k$	0.00775	0.01616
$\hat{\sigma}_k / \hat{\sigma}_1$	1.000	2.085
π_k^*	0.788	0.212
p_{kk}	0.9823	0.9342
d_k (days)	56.5	15.2
d_k (weeks)	≈ 11	≈ 3

Notes: The ratio $\hat{\sigma}_2/\hat{\sigma}_1 = 2.085$ indicates that the stress regime exhibits return volatility roughly double that of the stable regime. The stationary distribution π^* implies the system spends 78.8% of the time in regime 1 and 21.2% in regime 2. The average breakdown duration ($d_2 = 15.2$ days) is roughly 3.7 times shorter than the average duration of stable periods ($d_1 = 56.5$ days), confirming the empirically documented asymmetry between prolonged stable phases and brief but intense stress episodes.

9 Regime-Switching Models in Finance

Markov-switching models, introduced by Hamilton [**hamilton1989**] for U.S. business cycle analysis, apply HMM structures within econometric time series. Hamilton demonstrated that GDP dynamics could be better described by two regimes with distinct statistical characteristics (expansion and recession) governed by a Markov chain, rather than by constant-parameter linear models. Since then, the Markov-switching class of models has been applied to model returns, volatility, interest rates, and correlations.

Hamilton's Model

Hamilton's original model specifies:

$$r_t = \mu_{s_t} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2_{s_t}) \tag{69}$$

where s_t follows a Markov chain on $\{1, 2\}$. The unconditional distribution is a Gaussian mixture:

$$p(r_t) = \pi^*_1 \mathcal{N}(\mu_1, \sigma^2_1) + \pi^*_2 \mathcal{N}(\mu_2, \sigma^2_2) \tag{70}$$

with $\boldsymbol{\pi}^*$ given by equation (51). This model justifies two stylized facts: non-Gaussianity (leptokurtic unconditional distribution emerges as a Gaussian mixture) and regime shifting (variation in s_t describes structural process changes). Ang and Bekaert [**ang_bekaert2002**] document that two-regime Markov-switching models produce a better fit for equity and bond returns than constant parameter models; in particular, they capture regime-dependent correlations well.

Markov-Switching VAR

Hamilton's model generalizes to the multivariate context via the Markov-Switching VAR (MS-VAR):

$$\boldsymbol{y}_t = \boldsymbol{\mu}_{s_t} + \sum_{j=1}^{P} \boldsymbol{\Phi}_{j,s_t} \boldsymbol{y}_{t-j} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{s_t}) \tag{71}$$

In this model, the conditional mean, autoregressive coefficients, and covariance matrix all change according to the current regime s_t. MS-VAR is particularly relevant for multivariate pairs trading: the regime-dependent correlation matrix $\boldsymbol{\Sigma}_{s_t}$ captures the empirically documented correlation breakdown, enabling the distinction between a high-correlation/stable cointegration regime and an unstable correlation/degraded cointegration regime.

Hamilton Filter

This is the specialization of the forward recursion (56) to the econometric context. Let $\hat{\boldsymbol{\xi}}_{t|t-1}$ denote the $K \times 1$ vector of predicted regime probabilities and $\hat{\boldsymbol{\xi}}_{t|t}$ the vector of filtered probabilities:

$$\hat{\boldsymbol{\xi}}_{t|t-1} = \mathbf{P}' \, \hat{\boldsymbol{\xi}}_{t-1|t-1} \tag{72}$$

$$\hat{\xi}^{(k)}_{t|t} = \frac{f_k(\boldsymbol{y}_t) \, \hat{\xi}^{(k)}_{t|t-1}}{\sum_{j=1}^{K} f_j(\boldsymbol{y}_t) \, \hat{\xi}^{(j)}_{t|t-1}} \tag{73}$$

Equation (72) propagates regime probabilities through the transition matrix, while (73) updates probabilities by applying Bayes' theorem. The numerator is the product of the likelihood of the observation in regime k and the predicted probability of regime k, and the denominator is the normalization constant. The structure is identical to Bayesian updating (37) applied to the discrete state space.

10 Market Regime Detection

Regime detection is the problem of inferring, given the estimated model $\hat{\boldsymbol{\lambda}}$ and an observation sequence $\boldsymbol{y}_{1:t}$, the current market regime s_t. In pairs trading, this inference is operational meaning the filtered probability of being in a favorable regime dictates whether a KF-generated signal is executed or ignored.

The three main quantities of interest for regime detection are:

1. **Filtered Probabilities:** $\hat{\xi}_{t|t}^{(k)} = \Pr(s_t = k \mid \boldsymbol{y}_{1:t}, \hat{\boldsymbol{\lambda}})$. These rely solely on information up to time t and are therefore the only ones usable for real-time trading decisions. They are calculated via recursion (72)–(73).

2. **Smoothed Probabilities:** $\hat{\xi}_{t|T}^{(k)} = \Pr(s_t = k \mid \boldsymbol{y}_{1:T}, \hat{\boldsymbol{\lambda}})$. These utilize the entire time series to yield more accurate regime estimates at time t, but require future observations. They can only be used for ex-post analysis, structural break identification, and model validation.

3. **Viterbi Sequence:** $\hat{s}_{1:T}^{*}$. The globally most probable sequence of latent states, calculated using the Viterbi algorithm. It provides a sharp segmentation of the time series into regimes, useful for visualization and regime-dependent parameter estimation.

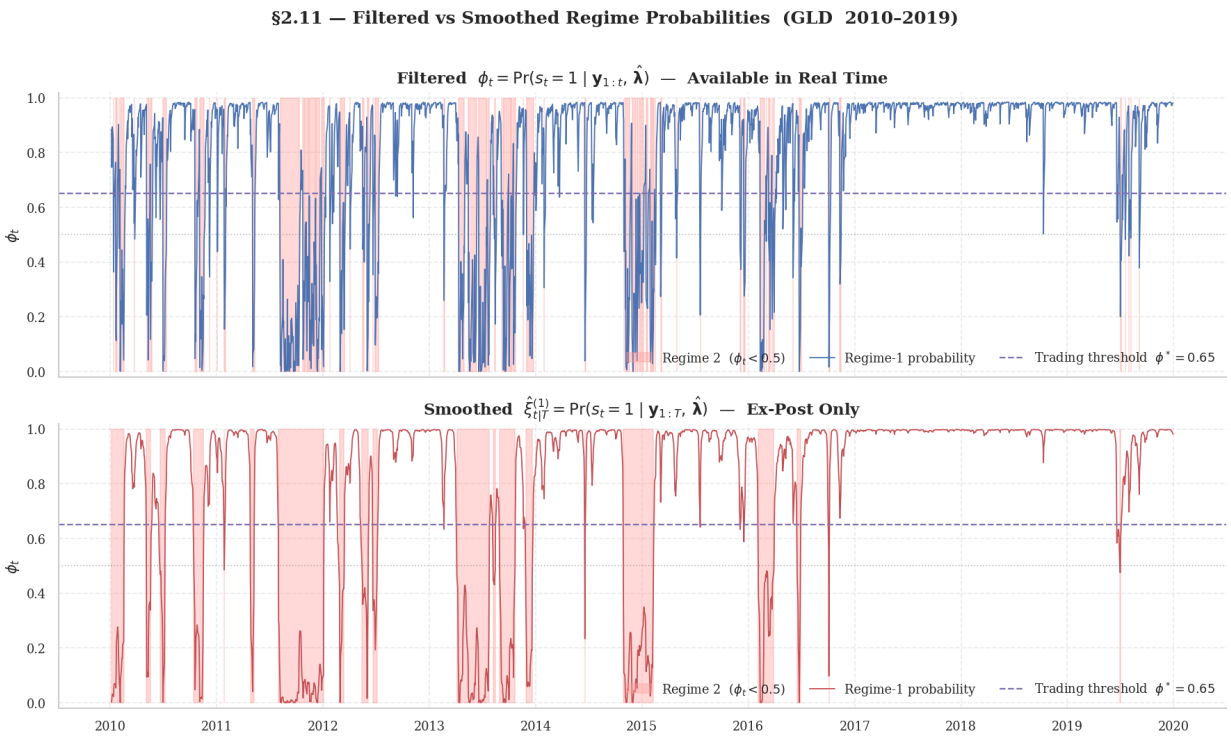


Figure 5: Filtered regime probabilities $\phi_t = \Pr(s_t = 1 \mid \boldsymbol{y}_{1:t}, \hat{\boldsymbol{\lambda}})$ (top panel) and smoothed probabilities $\hat{\xi}_{t|T}^{(1)} = \Pr(s_t = 1 \mid \boldsymbol{y}_{1:T}, \hat{\boldsymbol{\lambda}})$ (bottom panel) on GLD (2010–2019). Shaded areas indicate periods when $\phi_t < 0.5$ (regime 2 prevailing). The dashed line is the operational threshold $\phi^* = 0.65$. The physiological lag of filtered probabilities relative to smoothed ones at regime transitions highlights the informational cost of real-time operation versus ex-post analysis.

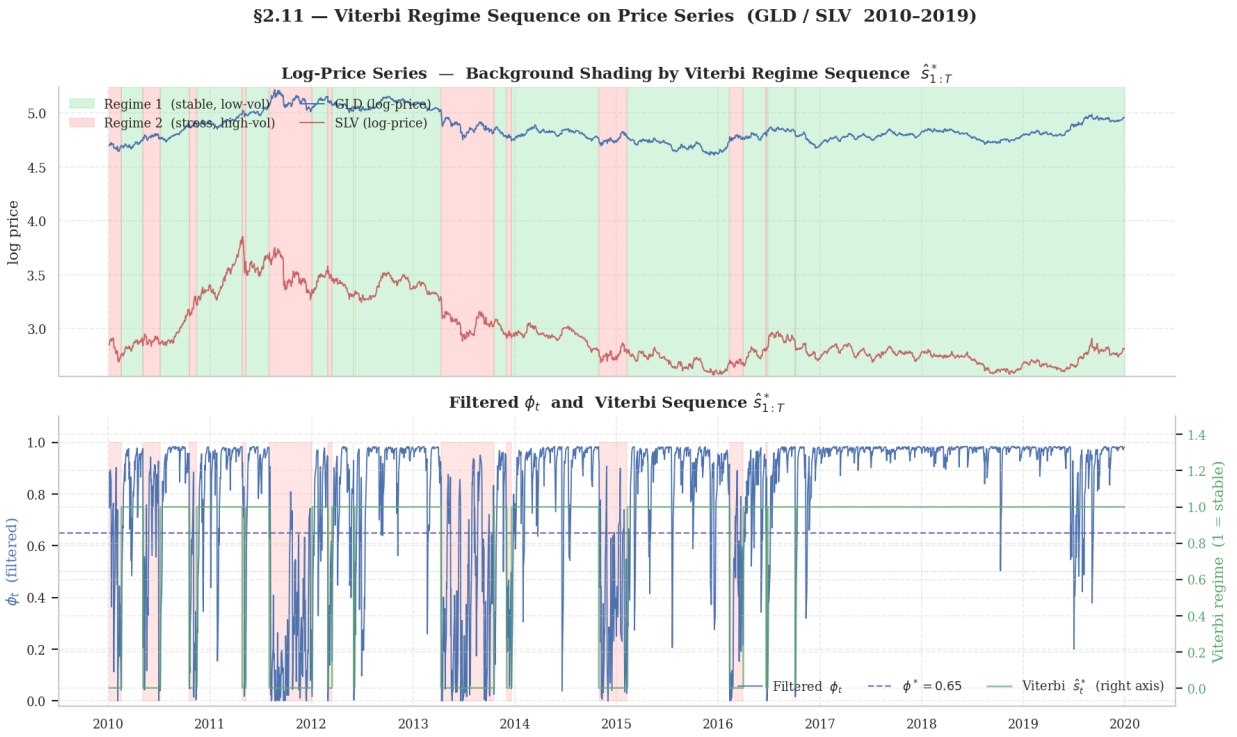


Figure 6: Top panel: time series of GLD and SLV log-prices (2010–2019) with background colored by the Viterbi sequence $\hat{s}^*_{1:T}$ (green = stable regime 1, pink = stress regime 2). Bottom panel: filtered probabilities ϕ_t (left axis) and the Viterbi sequence displayed as a step function (right axis). The correspondence between regime 2 periods and known market stress events (Gold ATH 2011–2012, China shock 2015–2016, 2018 sell-off) confirms the HMM's ability to structurally identify degraded cointegration periods.

The Regime Number Hyperparameter

The number of regimes K is a model hyperparameter and should be selected balancing data fit against parsimony. Standard information criteria are computed on the log-likelihood optimized by Baum-Welch. For an HMM with univariate Gaussian emissions, the number of free parameters is given by:

$$d_K = \underbrace{K(K-1)}_{\text{transitions}} + \underbrace{2K}_{\text{means and variances}} = K^2 + K \tag{74}$$

Information criteria are:

$$\text{AIC} = -2\log\mathcal{L}(\hat{\boldsymbol{\lambda}}) + 2d_K \tag{75}$$

$$\text{BIC} = -2\log\mathcal{L}(\hat{\boldsymbol{\lambda}}) + d_K \log T \tag{76}$$

BIC penalizes complexity more strongly and is preferred for HMMs. Practically, for pairs trading, $K = 2$ (low/high volatility) is almost always sufficient and produces interpretable regimes. Models with $K = 3$ can capture a third intermediate phase (medium volatility regime) but at the cost of increased numerical instability.

Trading Indicators

The filtered probability of being in a favorable regime is:

$$\phi_t = \hat{\xi}^{(1)}_{t|t} = \Pr(s_t = 1 \mid \boldsymbol{y}_{1:t}, \hat{\boldsymbol{\lambda}}) \tag{77}$$

where regime 1 is conventionally the low volatility and stable cointegration regime (identified during estimation as the regime with $\hat{\sigma}_1 < \hat{\sigma}_2$ and $\hat{\mu}_1 > \hat{\mu}_2$ for pair spreads with a positive bias). A

composite indicator combining regime probability with estimated duration in the current regime is:

$$\Phi_t = \phi_t \cdot \left(1 - e^{-\phi_t/(1-p_{11})}\right) \tag{78}$$

where the second factor penalizes recent transitions (when ϕ_t has just risen from zero, the system might still be in a transient phase). A simple threshold $\phi_t > \phi^*$ is typically used, optimized on the training sample.

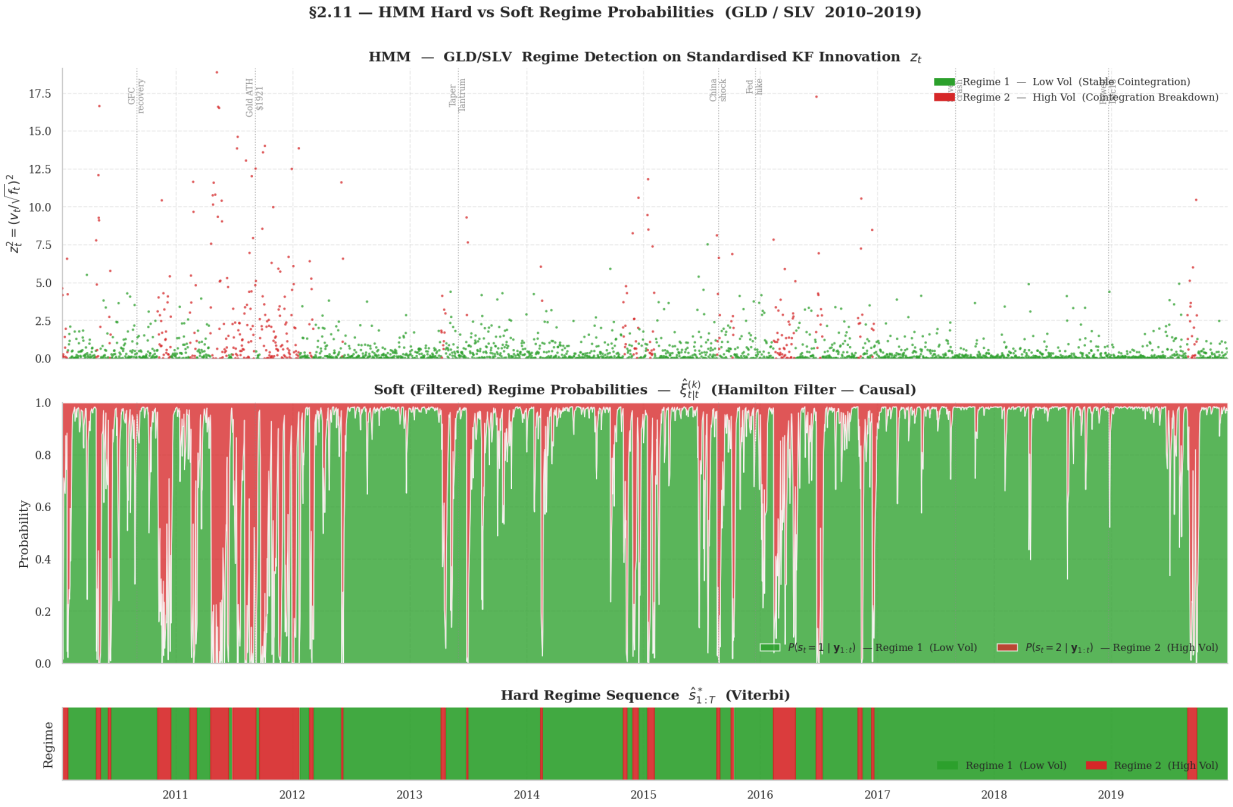


Figure 7: HMM regime detection on GLD/SLV (2010–2019). Top panel: scatter plot of $z_t^2 = (v_t/\sqrt{f_t})^2$ colored by Viterbi regime (green = regime 1, red = regime 2); vertical dashed lines indicate relevant market events. Middle panel: *soft* filtered regime probabilities $\hat{\xi}_{t|t}^{(k)}$ (causal Hamilton filter) as a stacked area chart — the green component represents $\Pr(s_t = 1 \mid \boldsymbol{y}_{1:t})$ and the red one $\Pr(s_t = 2 \mid \boldsymbol{y}_{1:t})$. Bottom panel: *hard* regime sequence $\hat{s}_{1:T}^{*}$ (Viterbi algorithm). Comparing the middle and bottom panels highlights the structural difference between continuous probabilistic estimation (soft) and optimal discrete classification (hard): the former captures uncertainty at regime transitions, the latter provides a sharp segmentation usable for regime-dependent parameter estimation.

Table 3 reports the parameters of the HMM estimated on the z_t^{KF} z-score utilized for regime detection in Figure 7. Note that these parameters are not comparable to those in Table 2, as the input is the standardized z-score rather than raw log-returns.

Table 3: Estimated parameters for the two-state HMM on the adaptive z-score $z_t^{\mathrm{KF}} = v_t/\sqrt{f_t}$ of GLD/SLV (2010–2019). Compared to Table 2, the input here is the standardized z-score rather than raw log-returns; hence, the parameters operate on a different scale and are not directly comparable.

Parameter	**Regime 1 (Low Vol)**	**Regime 2 (High Vol)**
$\hat{\mu}_k$	-0.00965	-0.04283
$\hat{\sigma}_k$	0.68765	1.79467
$\hat{\sigma}_k/\hat{\sigma}_1$	1.000	2.609
π_k^*	0.811	0.189
p_{kk}	0.9763	0.8979
d_k (days)	42.2	9.8
d_k (weeks)	≈ 8	≈ 2

11 Integrating HMM and KF in Pairs Trading

Integrating the Kalman Filter and Hidden Markov Models into a pairs trading architecture constitutes a key contribution. The two models play complementary and non-overlapping roles in the signal generation chain. The Kalman Filter estimates the cointegrating relationship structure and generates the quantitative signal, while the HMM evaluates market context quality and acts as a qualitative filter on signal execution. This separation is intentional and reflects a precise view of the decision-making process: it is not enough for the spread to be far from the mean; the market regime must also be compatible with mean reversion.

Table 4: Comparison among specifications of the signal generation framework for pairs trading. The KF-HMM framework resolves all three biases identified in equation (10).

Property	**OLS**	**KF**	**KF-HMM-GARCH**
Stationary numerator	No	Yes	Yes
Adaptive denominator	No	Partial	Yes
Corrects parameter drift	No	Yes	Yes
Corrects volatility clustering	No	No	Yes
Operational regime filter	No	No	Yes
Suppresses signals in regime 2	No	No	Yes
Requires periodic re-estimation	Yes	No	No
Causal (real-time)	Yes	Yes	Yes
Parameters to optimize	0	3	6

Notes: The optimization parameters for the KF-HMM-GARCH framework are $(\hat{\sigma}_\varepsilon^2, \hat{q}_\alpha, \hat{q}_\beta)$ for the KF, estimated via MLE, plus the operational thresholds $(z^*, z^{\mathrm{exit}}, \phi^*)$ optimized via grid search during backtesting. GARCH parameters $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$ are estimated via MLE on KF residuals and do not require separate optimization.

Trading System Architecture

The integrated KF-HMM system operates via the following pipeline, repeated at each new available data point:

1. **KF Update:** Given the new price pair $(Y_{1,t}, Y_{2,t})$, the Kalman Filter prediction and update steps are executed, yielding the updated filtered estimate $\hat{\boldsymbol{\alpha}}_{t|t} = (\hat{\alpha}_{t|t}, \hat{\beta}_{t|t})'$, the innovation v_t, and its variance f_t.

2. **HMM Update:** Using the current observation $\boldsymbol{y}_t$ (e.g., standardized returns or normalized z-score from the previous step), the filtered regime probability ϕ_t is updated via the Hamilton

filter (72)–(73).

3. **Dynamic Z-Score Calculation:** The KF innovation, normalized by its conditional standard deviation, produces the adaptive z-score defined in (47):

$$z_t = \frac{v_t}{\sqrt{f_t}} \tag{79}$$

This z-score is already adjusted for current parametric uncertainty. To also correct for spread volatility clustering (which f_t only partially captures, since it addresses parametric uncertainty), $\sqrt{f_t}$ is replaced by $\sqrt{f_t + \hat{\sigma}^2_{S,t}}$, where $\hat{\sigma}^2_{S,t}$ is the GARCH conditional variance computed on the filtered KF residuals. This specification is often adopted in empirical applications.

4. **Regime-Dependent Signal Generation:** The trading signal τ_t combines the dynamic z-score with the regime probability:

$$\tau_t = \begin{cases} +1 & \text{if } z_t < -z^* \text{ and } \phi_t > \phi^* \quad \text{(long spread: long } Y_1\text{, short } Y_2\text{)} \\ -1 & \text{if } z_t > +z^* \text{ and } \phi_t > \phi^* \quad \text{(short spread: short } Y_1\text{, long } Y_2\text{)} \\ 0 & \text{otherwise} \quad \text{(flat or close position)} \end{cases} \tag{80}$$

where z^* is the entry threshold, and the position is closed when $|z_t|$ falls below an exit threshold $z^{\text{exit}} < z^*$.

5. **Position Sizing:** The position size can be modulated depending on the regime probability and z-score magnitude:

$$w_t = \bar{w}\, \phi_t \cdot \min\left(\frac{|z_t|}{z^*},\, w_{\max} \right) \tag{81}$$

where $\bar{w}$ is the maximum position size and $w_{\max}$ is a maximum multiplier. This ensures positions are larger when the z-score is wide and the regime is favorable, and automatically shrink during regime uncertainty periods.

The condition $\phi_t > \phi^*$ in rule (80) is not an arbitrary choice but reflects an econometric reality. The signal $z_t > z^*$ indicates the spread is significantly above its historical average. This deviation can have two distinct causes:

1. The spread has temporarily departed from long-term equilibrium and will revert to it. This is a genuine mean reversion opportunity. Such a situation features $\phi_t \approx 1$ (stable cointegration regime).

2. The cointegrating relationship has broken down. This is a structural break, not a temporary deviation. Such a situation features $\phi_t \approx 0$ (degraded cointegration regime).

Without the regime condition, the signal $z_t > z^*$ is ambiguous. The condition $\phi_t > \phi^*$ selects the first case: the signal is executed only when the HMM is sufficiently confident that cointegration is stable. The optimal threshold ϕ^* is determined empirically based on the trade-off between the number of generated signals and selection quality (typical values fall within the $[0.6, 0.8]$ interval).

Sequential Estimation of the System

The KF-HMM architecture is estimated in two independent steps:
Step 1 (KF): Hyper-structural parameters $\hat{\boldsymbol{\psi}} = (\hat{\sigma}^2_\varepsilon, \hat{q}_\alpha, \hat{q}_\beta)$ are estimated by maximizing log-likelihood (33) via L-BFGS-B. This yields time series of filtered estimates $\{\hat{\boldsymbol{\alpha}}_{t|t}\}_{t=1}^T$, innovations $\{v_t\}$, and innovation variances $\{f_t\}$. Dynamic spread $S_t = v_t$ and adaptive z-score $z_t = v_t/\sqrt{f_t}$ are calculated.

Step 2 (HMM): The HMM with K regimes is estimated on regime observations $\boldsymbol{y}_t^{\text{HMM}}$, which can be gross returns, standardized returns, or z-scores. Parameters $\hat{\boldsymbol{\lambda}} = (\hat{\mathbf{P}}, \hat{\boldsymbol{\pi}}, \{\hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k\}_{k=1}^K)$ are estimated via the Baum-Welch algorithm with $N_{\text{init}} = 50$ random initializations. This yields the filtered probabilities series $\{\phi_t\}_{t=1}^T$.
Step 3 (Threshold Optimization): Thresholds $(z^*, z^{\text{exit}}, \phi^*)$ are optimized on a training sample by maximizing a performance metric (e.g., Sharpe ratio) using a three-dimensional grid search, followed by walk-forward validation to avoid overfitting [**lopezdeprado2018**].

Comparison with Alternative Approaches

The simplest approach (z-score calculated with rolling mean and rolling standard deviation over fixed windows) is computationally immediate but non-adaptive: the window length is a fixed hyperparameter that ignores market variations. A window that is too short yields noisy estimates, while a window that is too long yields lazy estimates failing to adapt to structural breaks.
Pole's model [**pole2007**] suggests a Kalman Filter for the moving average alone, leaving standard deviation constant. Our architecture extends this approach by simultaneously estimating the mean, hedge ratio, and conditional variance, and adding the market regime dimension via HMM.
Liu et al. [**liu2017**] combine KF for the mean and GARCH for variance, but do not incorporate regime detection as a signal execution filter. The addition of the HMM is a specific methodological contribution in this framework: the condition $\phi_t > \phi^*$ systematically reduces false signals during structural breaks, as documented empirically.

This document defined the foundations of the dynamic state-space framework. The Kalman Filter and Hidden Markov Models are adopted as complementary tools operating on two distinct levels: the continuous level of hedge ratio variation and the discrete level of market regime variation. The integration of both into a unified architecture (where regime detection acts as a qualitative filter over the quantitative KF signal) represents a major methodological proposal.



Figure 8: Comparison among three z-score specifications on GLD/SLV (2010–2019). Top panel: static z-score $z_t^{\text{OLS}} = S_t^{\text{OLS}} / \hat{\sigma}_\varepsilon$ with a fixed denominator. Middle panel: adaptive z-score $z_t^{\text{KF}} = v_t / \sqrt{f_t}$ adjusting solely for the Kalman Filter's parametric uncertainty. Bottom panel: full-denominator z-score $z_t^{\text{KF+GARCH}} = v_t / \sqrt{f_t + \hat{\sigma}^2_{S,t}}$, where $\hat{\sigma}^2_{S,t}$ is the GARCH(1,1) conditional variance estimated on innovations v_t; the right axis displays the two denominator components ($\sqrt{f_t}$ in green and $\hat{\sigma}_{S,t}$ in light blue, both $\times 10^3$). This demonstrates that moving from OLS to KF resolves systematic numerator drift, rendering the z-score stationary by design. Adding the GARCH component to the denominator ($\hat{\sigma}^2_{S,t}$, bottom panel, right axis) further widens the denominator during conditional spread volatility spikes (2011–2012, 2015–2016), correcting the second bias source discussed in (10). Empirical validation of the combined denominator's superiority versus $\sqrt{f_t}$ alone is deferred to backtesting analysis.

Table 5: Estimated GARCH(1,1) model parameters on Kalman Filter innovations v_t on GLD/SLV (2010–2019). The model is estimated on $v_t \times 10^4$ for numerical stability, and parameters are scaled back to original units. Persistence $\hat{\alpha} + \hat{\beta} = 0.986$ implies highly persistent conditional spread volatility.

Parameter	**Description**	**Estimated Value**
$\hat{\omega}$	Constant term	1.448×10^2
$\hat{\alpha}$	ARCH coefficient (squared shocks)	0.0817
$\hat{\beta}$	GARCH coefficient (lagged variance)	0.9043
$\hat{\alpha} + \hat{\beta}$	Conditional volatility persistence	0.9861
$\hat{\sigma}^2_\infty$	Unconditional variance $\hat{\omega}/(1 - \hat{\alpha} - \hat{\beta})$	1.034×10^4

Table 6: Statistical comparison of the three z-score specifications on GLD/SLV (2010–2019). Stationarity is assessed visually via range and standard deviation: a properly standardized z-score exhibits std ≈ 1. The KF+GARCH column shows amplitude compression due to a larger denominator during high conditional volatility periods.

Statistic	z_t^{OLS}	z_t^{KF}	$z_t^{\text{KF+GARCH}}$
Min	-2.62	-7.38	-3.86
Max	$+3.83$	$+4.34$	$+2.44$
Std	1.000	1.001	0.658
Denominator	Fixed $\hat{\sigma}_\varepsilon$	Adaptive $\sqrt{f_t}$	$\sqrt{f_t + \hat{\sigma}^2_{S,t}}$
Systematic drift	Yes (2013–2019)	No	No
Corrects param drift	No	Yes	Yes
Corrects vol clustering	No	No	Yes

Notes: The standard deviation of $z_t^{\text{KF+GARCH}} = 0.658 < 1$ indicates the combined denominator $\sqrt{f_t + \hat{\sigma}^2_{S,t}}$ over-standardizes the spread in high-volatility periods, compressing signals more than necessary. This suggests the threshold z^* should be re-calibrated for the KF+GARCH denominator during backtesting, typically lowering it from $z^* = 2$ to $z^* \approx 1.3$.

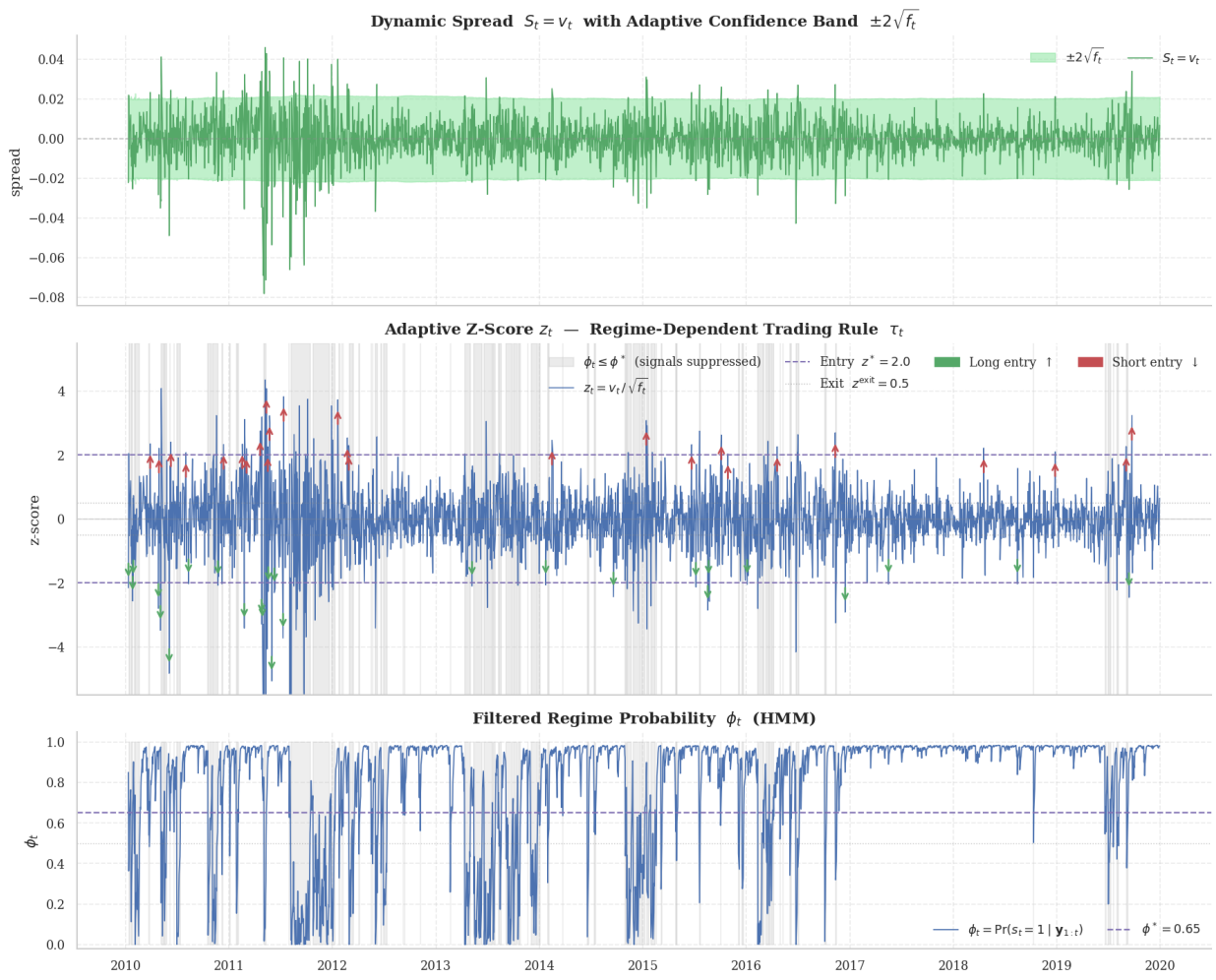


Figure 9: Operational dashboard of the KF-HMM system on GLD/SLV (2010–2019). Top panel: dynamic spread $S_t = v_t$ alongside adaptive confidence band $\pm 2\sqrt{f_t}$. Middle panel: adaptive z-score (No GARCH) $z_t = v_t/\sqrt{f_t}$ with entry thresholds $\pm z^* = 2$ and exit thresholds $\pm z^{\text{exit}} = 0.5$; green arrows denote long signals and red arrows short signals; grey areas highlight periods where $\phi_t \le \phi^* = 0.65$ and signals are suppressed by the regime filter. Bottom panel: filtered regime probability ϕ_t with threshold ϕ^*. Signal suppression during regime 2 periods — particularly during the 2015–2016 China shock and the 2018 sell-off — minimizes spurious trades executed amidst degraded cointegration phases.